Fingerprint counts

Fingerprints Up Close

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little Wonderer counts 38 ridges on one print and 52 ridges on another. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

38 ○ 52

Show your thinking.

2. Next, Big Wonderer presses 71 prints on one page and 67 prints on the next page. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

71 ○ 67

Show your thinking.

3. The little one finds 45 pores on one strip of tape and 45 pores on another. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

45 ○ 45

Show your thinking.

Fingerprint counts

Fingerprints Up Close

Name _____ Date _____

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4. The big one counts 34 ridge endings on a big drawing. What is 34 rounded to the nearest ten?

34 → nearest ten: ___

Show your thinking.

5. The little one draws 86 tiny pore dots. What is 86 rounded to the nearest ten?

86 → nearest ten: ___

Show your thinking.

6. The big one moves 65 wet marbles in a long game. What is 65 rounded to the nearest ten?

65 → nearest ten: ___

Show your thinking.

A hidden print

Tiny ridges carry sweat and oil. A touch copies their pattern, even when you cannot see it.

Three patterns

Loops curve back. Whorls make circles. Arches flow across with a bump. No two have ever been found to match.

Made before birth

Prints form before birth. Each fingertip grows in its own tiny way, so even twins differ.

Koala prints

Koalas and people both grip things. Scientists think ridges may help us feel fine textures.

A safe print lift

Detective tools are for trained lab workers. A pencil patch and clear tape make a safe print.

Wrinkly fingers

Nerves make bath fingers wrinkle. Maybe wrinkles help grip wet things. Scientists are still checking.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

Fingerprints Up Close

Explain a fingerprint clue in your words

Fingerprints Up Close · Curiosity day

Name _____ Date _____

Choose one picture. Say your idea first, then draw or write it.



Three patterns



Ridges up close



Wet marbles

Draw your own fingerprint big. Mark one place where a ridge ends or splits.

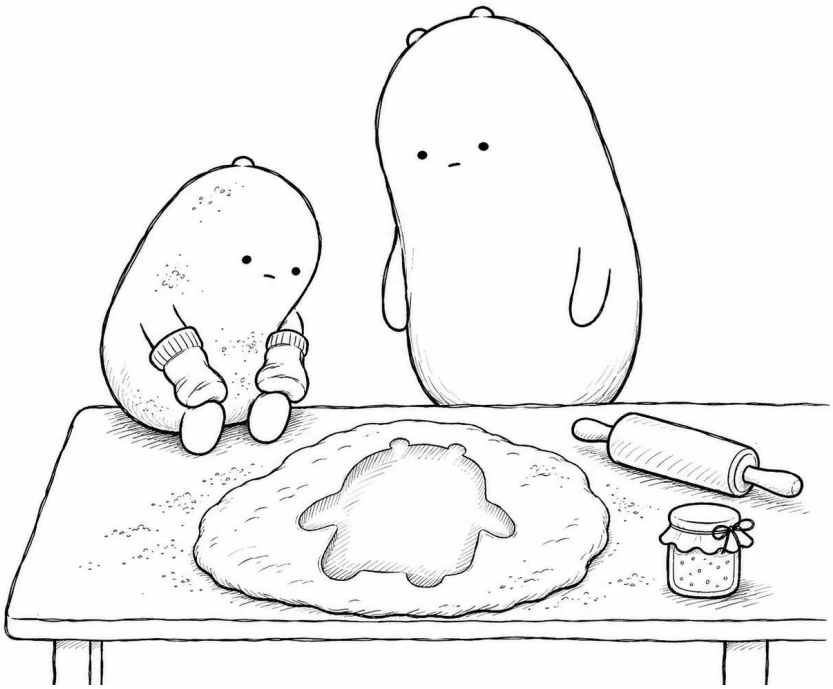
ridge · loop · whorl · arch · because · maybe

My print is a loop, whorl or arch because … how do the ridges flow?

Twins have different prints because … how does each fingertip grow?

Wrinkly fingers might help because … what might they grip?

Does your sentence tell what you noticed and use because or maybe to explain?



Little Wonderer Leaves Clues

Little Wonderer touched the jam jar lid,
the window, the doorknob, the drum. Yes, it did.
It patted the mirror, it poked at the cake,
it tapped every spoon just for tapping's sake.

"Oh no!" it cried out. "There are clues everywhere!
A loop on the cupboard! A whorl on the chair!
A detective could follow my trail through the house.
I'm as easy to find as a jam-covered mouse!"

Big Wonderer nodded and wiped off a pane.
"Your swirls will grow bigger but mostly remain.
No two have been found to match, pair after pair,
so a print is a clue that says someone was there."

So the little one wore socks on both hands all day,
to tiptoe through mischief and sneak right away.
But it tripped on the rug and it fell in the dough,
and now there's a Little-shaped print, head to toe.

Fingerprints Up Close

01 Is your print a loop, a whorl or an arch?



A loop, a whorl and an arch.

Look closer. Find the three prints. One has ridges that curve back, one has a circle, and one has a small hill in the middle.

A little explanation. A loop has ridges that come in and curve back out the same side. A whorl has circles or a spiral. An arch has ridges that flow across with a bump in the middle. Loops are the most common, and arches are the rarest.

Try this. Rub a pencil hard on paper to make a dark gray patch. Press one fingertip on it. Press clear tape on your fingertip, peel it off, and stick it on white paper. Is your print a loop, a whorl or an arch?

02 What tiny details are inside a print?



Up close, ridges end, split and carry tiny pores.

Look closer. Find a ridge that stops, and a ridge that splits in two. Look for the tiny dots along the top of the ridges.

A little explanation. Sweat pores sit in a row along the top of each ridge. Experts look at minutiae, the tiny places where a ridge ends or splits in two. A split is called a bifurcation. They compare these points, not just the pattern. Very rarely, a change in a gene called SMARCAD1 means a person is born with no fingerprints at all.

Try this. Look at your tape print with a magnifying glass if you have one. Find one place where a ridge ends. Then find one place where a ridge splits in two.

03 Why does a koala have fingerprints?



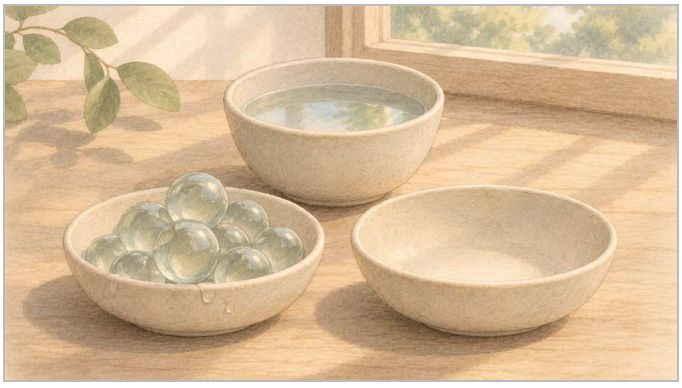
Koalas have prints that look very much like ours.

Look closer. Look at the koala's hand holding the branch. Find the tips of its fingers.

A little explanation. Koala prints look very much like ours. But koalas are only distantly related to us, and their cousins, wombats and kangaroos, do not have prints. Scientists think prints formed separately in both because both grip and handle things.

Try this. Draw a koala hand holding a branch. Then trace your own hand next to it. Tell one way they are alike.

04 Why do fingers get wrinkly in the bath?



Maybe wrinkly fingers help hold wet things.

Look closer. Find the two bowls, the wet marbles, and the bowl of water.

A little explanation. Wrinkly bath fingers are not just skin soaking up water. Your nerves control them. Tiny blood vessels under the skin get narrower, and the skin puckers. Scientists think wrinkles might help us grip wet things, but studies disagree, so for now we say maybe.

Try this. Time how fast you move wet marbles from one bowl to another with dry fingers. Soak your hands in warm water until they wrinkle, then try again. This is a fun try, not proof.